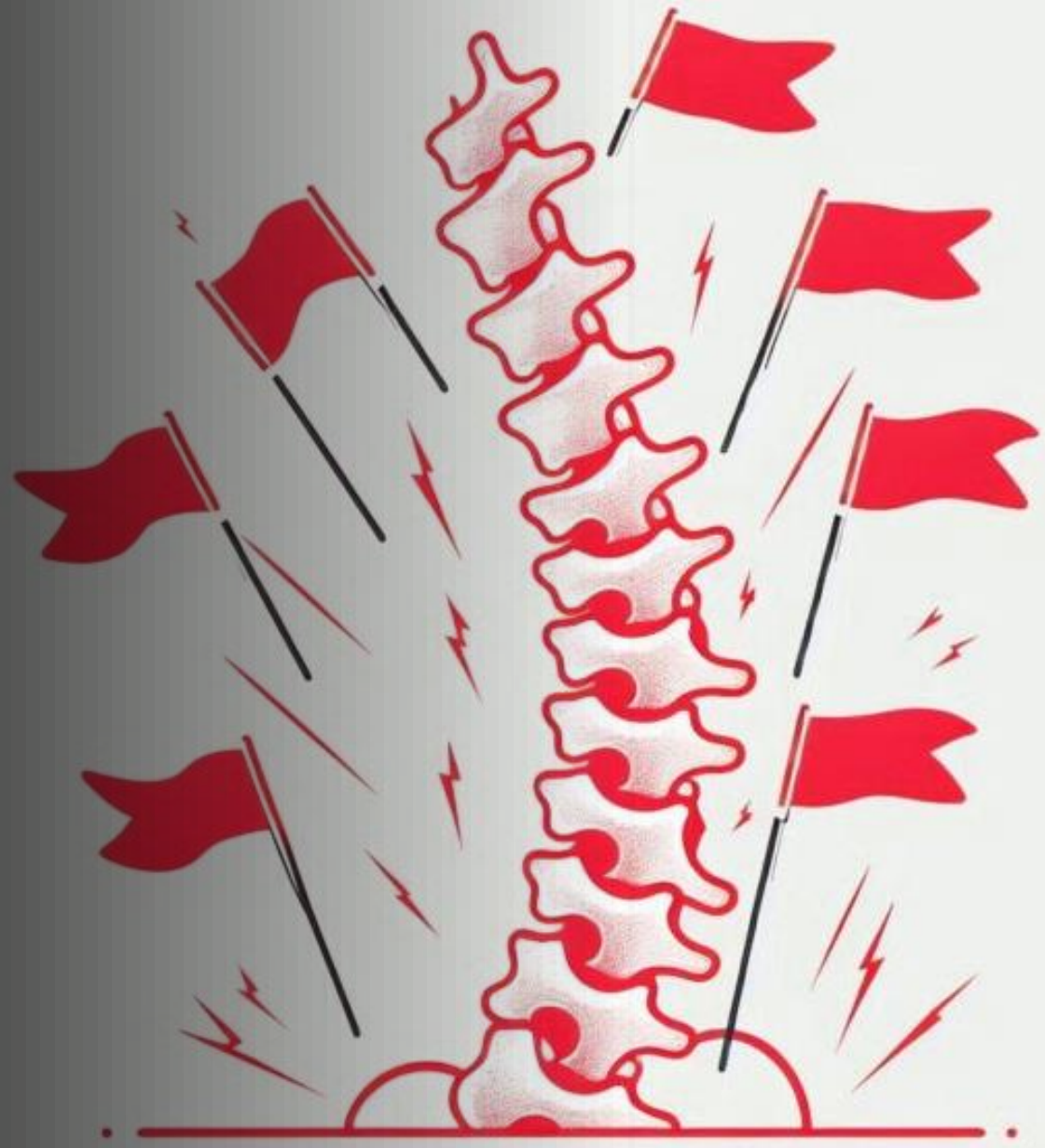
Spinal Pain Red Flags and masquerades

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Learning Outcomes

Serious pathologies that can masquerade as neuromusculoskeletal pain of **spinal** origin

The concept of red flags: signs and symptoms that can be indicative of serious pathology

The risk of affecting the blood supply to the brain when moving the neck

The current state of knowledge of this risk and how it can be minimised when treating patients with neck problems

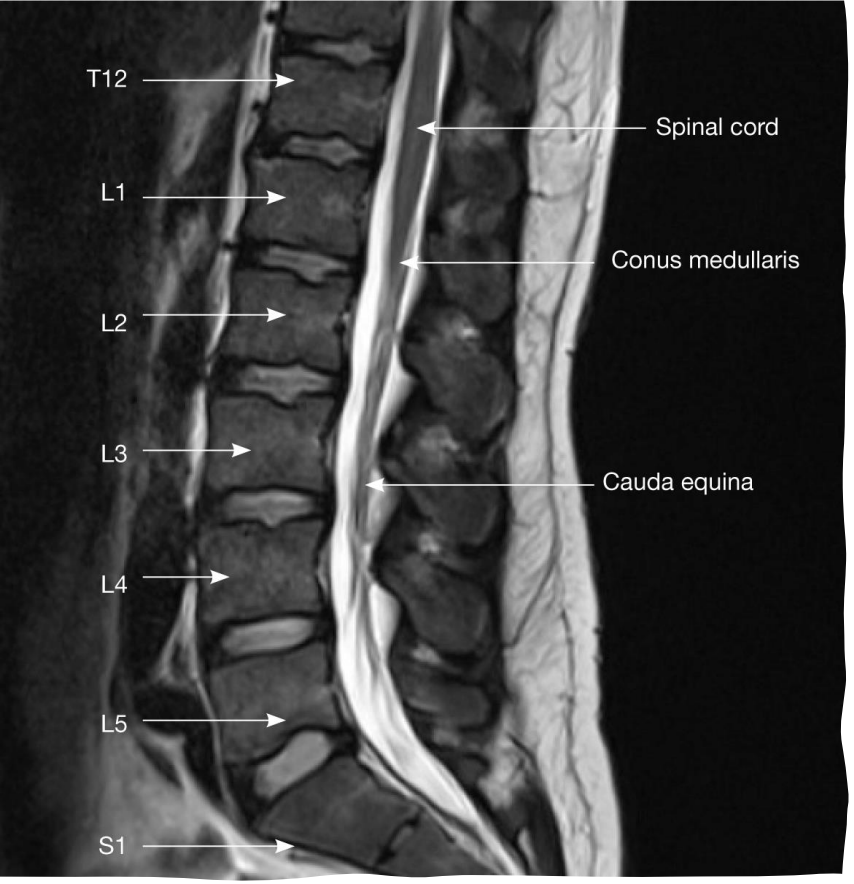
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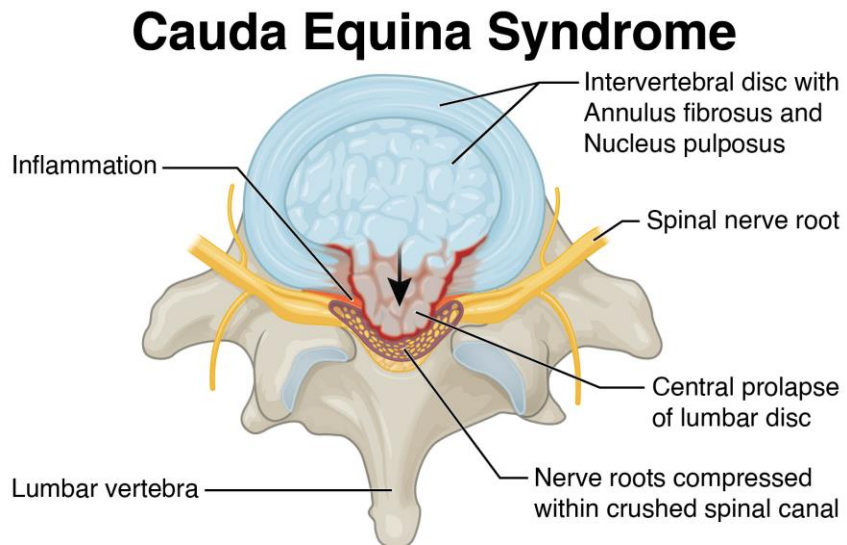
1. Concept of Red Flags
2. Cauda Equina Syndrome
3. Spinal Fractures
4. Spinal Malignancy
5. Spinal Infection
6. Cervical Myelopathy
7. Cervical arterial dysfunction (CAD)
8. Inflammatory Pathology
9. Visceral Pathology

Flag	Nature
Red	Signs of serious pathology
Orange	Psychiatric symptoms
Yellow	Beliefs, appraisals and judgements
	Emotional Responses
	Pain behaviour (including pain and coping strategies)
Blue	Perceptions about the relationship between work and health
Black	System or contextual obstacles

What are red flags?

- Red flags are specific attributes derived from a patient's medical history and the clinical examination that are usually linked with a high risk of having a serious disorder like an infection, cancer, or a fracture.
- Their main function is to raise the therapist's suspicion when they are combined·
- The flag system describes clinical and psychological flags and is comprised of **red, orange, yellow, blue and black.**





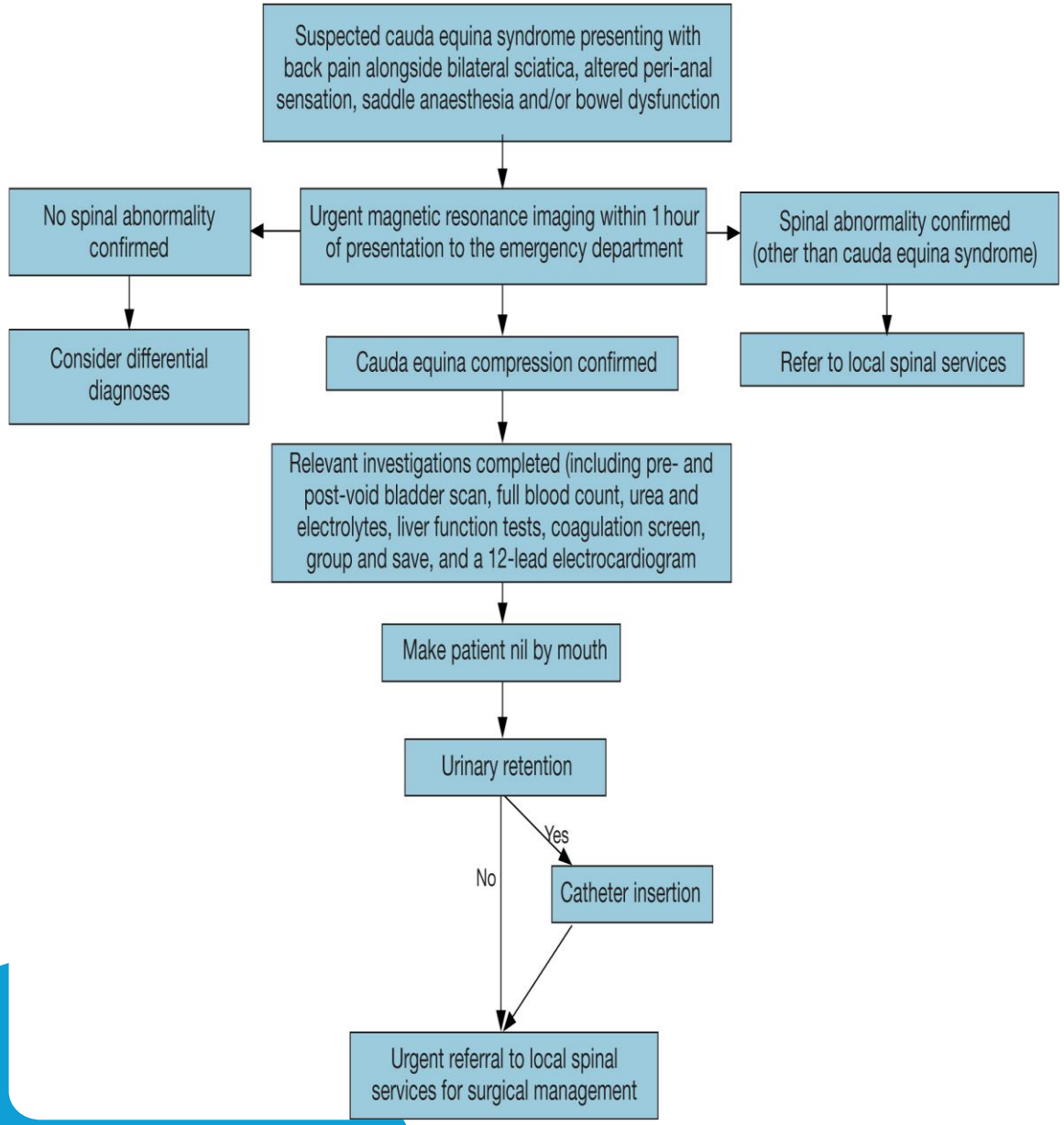
Cauda Equina Syndrome

- Cauda equina provides innervation to LL's, sphincter, sensory innervation to saddle and parasympathetic innervation to bladder + bowel
- CES first described 1934; still no universal definition
- *"a patient presenting with acute low back pain with a suggestion of a disturbance of their bladder or bowel function and/or saddle sensory disturbance should be suspected of having CES"*(Germon et al, 2015)
- Rare but life-altering condition if missed
- Estimated incidence (UK): 0.002% → 0.04% of all back pain → 0.4% of back pain seen in spine centre
- 1/5 patients will have a poor outcome, with on-going bladder, bowel and sexual dysfunction + signifiant psychosocial consequences (Gardner et al, 2011).



Causes of CES

Compressive causes	Lumbar disc herniation
	Lumbar spinal stenosis
	Spinal neoplasms
	Spinal arteriovenous malformations
	Epidural haematoma
Non-compressive causes	Spinal arachnoiditis
	Ischaemic insults
	Inflammatory conditions, eg ankylosing spondylitis
	Spinal infections



CES – 5 Key Features

1. Bilateral neurogenic sciatica
2. Reduced perianal sensation
3. Altered bladder function
4. Loss of anal tone
5. Sexual dysfunction

Four stages of CES (Todd and Dickson 2016)

CES STAGES	
CES Suspected	Bilateral radicular pain
CES Incomplete	Urinary difficulties of neurogenic origin (altered urinary sensation, loss of desire to void, poor urinary stream, need to strain to micturate)
CES Retention	Neurogenic urine retention (painless urinary retention with overflow incontinence)
CES Complete	Objective loss of cauda equina function, absent perineal sensation, patulous anus, paralyzed and insensate bladder/bowel

Subjective Ax

- Red flags as prev
- **+PMHx** (CES, Ca, masqueraders e.g. Diabetes, MS, Benign prostatic hyperplasia)

- **+ Medications**

✓ Cholinergics - voiding of urine

✓ Anticholinergics - retention of urine

✓ Opioids - constipation, reduced bladder sensation

✓ Anticonvulsants – urinary incontinence

✓ Antidepressants - Retention, sexual dysfunction, reduced awareness of need to

✓ pass urine

✓ NSAIDS - Retention twice as likely in men than women

Cauda Equina Syndrome

EXAMINATION

- Full neuro – dermatomes, myotomes, reflex's
- UMN tests (Babinski, Hoffmans, Clonus)
- Digital rectal examination (DRE)? – evidence weak, consider appropriate training + chaperone
- Consider testing sensation to light touch and pin prick throughout the saddle region including buttocks, inner thighs and perianal region (Todd and Dickson, 2016).

Making an urgent referral

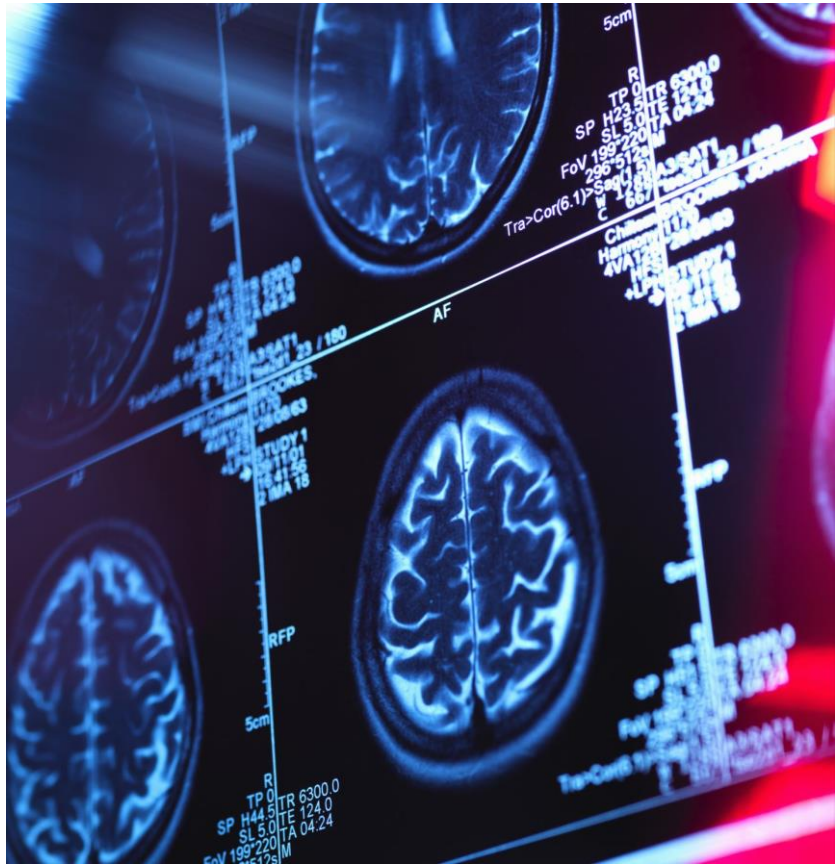
- The patient should be referred urgently (to be seen within two weeks) to the nearest MSK triage service if they present with sudden onset bilateral radicular leg pain (sciatica) or unilateral radicular leg pain that has progressed to bilateral without CES signs or symptoms.
- During the wait for an urgent appointment, it is important that the patient knows how to identify and act upon any deterioration in their presentation. ‘Safety Net’ the patient with a warning card and access to the MACP video (see Safety Netting).
- If a patient then experiences and reports deteriorating or new CES symptoms, make an emergency referral. If the symptoms are static, and remain so, the patient should continue with an urgent referral to the MSK triage service.
- Emergency referral after a telephone assessment identifying CES symptoms is acceptable if immediate face to face review is not possible or will delay referral.

Referral

1- Bladder scan: A bladder scan is a useful adjunct in the assessment of a patient with suspected CES. Bladder scans should NOT be used in isolation or as a discriminator in deciding to request an MRI or undertake emergency surgery.

2- MRI Imaging: MRI imaging is a critical diagnostic investigation in the management of patients with suspected CES. An emergency MRI for suspected CES should be undertaken as soon as possible, and certainly within four hours of request to radiology.

3- Surgical intervention.



When to refer to the spinal surgical team

Imaging outcome	Action required
Cauda Equina compression confirmed	Immediate referral to spinal surgical service Keep patient nil by mouth If transfer required after (response by surgical service) arrange Category 2 ambulance transfer
No Cauda Equina compression but neural compression to explain radicular pain (sciatica)	Safety net about progression of CES symptoms via video and card Advise patient that pain is very likely to improve Referral to MSK interface service/triage service
No cause for symptoms found	Consider alternative diagnoses Consider referral to other specialties

Documentation

Urgent referral documentation An urgent referral should document the following:

- flag referral as urgent;
- sudden onset bilateral radicular pain or unilateral radicular leg pain that has progressed to bilateral without CES symptoms or onset of CES symptoms >2weeks ago which are now static;
- signs and symptoms of bilateral sciatica and including duration, frequency and progression;
- details of assessment of patient;
- time and date of assessment(s);
- examination findings;
- physical examination of power and sensation in the lower limbs; 9
- a digital rectal examination is not necessary but subjective perianal sensation should be recorded.

Communication + Safety Netting

Safety netting Safety netting for patients experiencing back pain with other symptoms is crucial to ensuring patients know how and when to seek help at the appropriate time.

A video and warning card from the Musculoskeletal Association of Chartered Physiotherapists (MACP) has been developed for a patient audience.

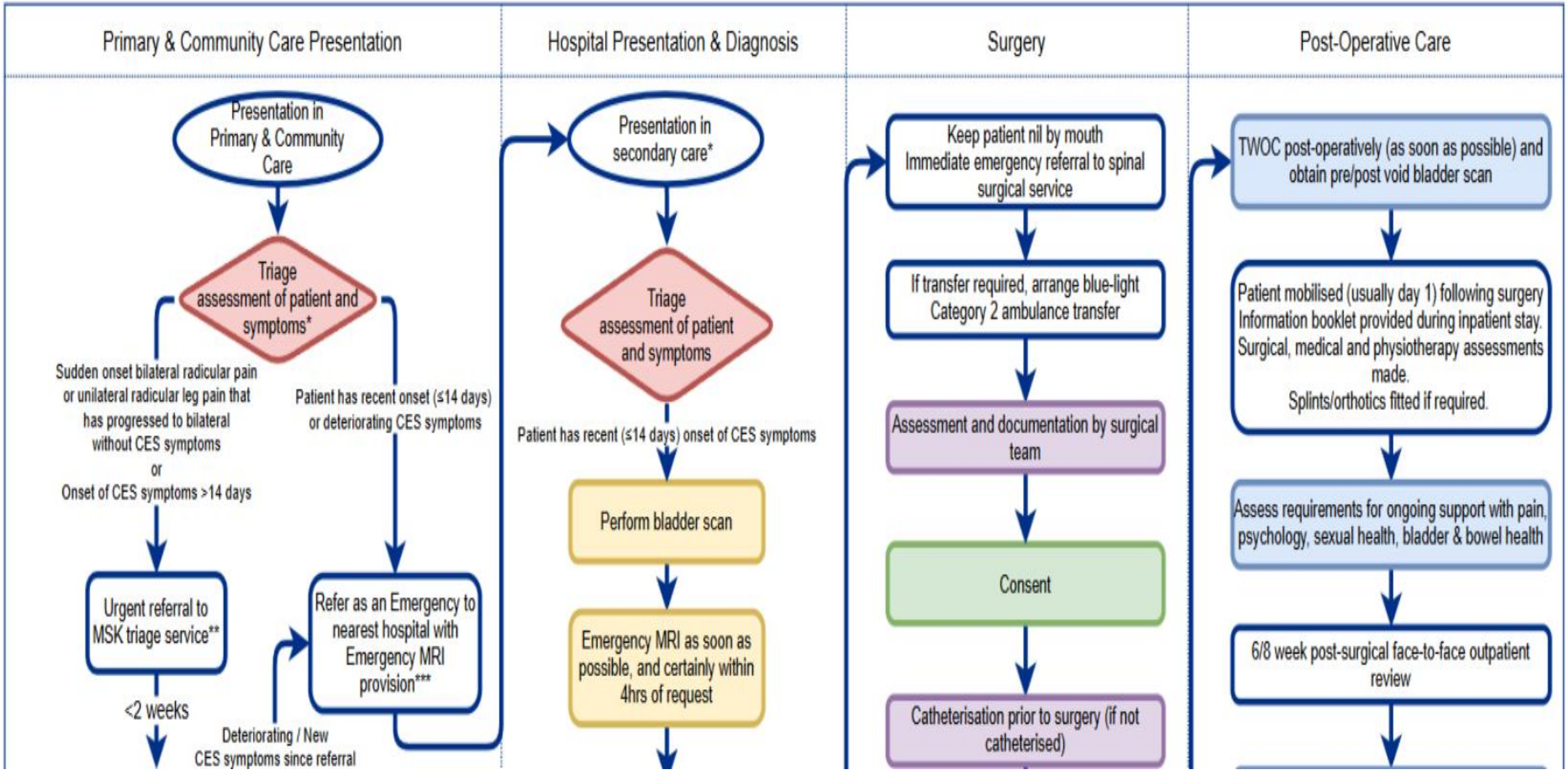
It is accessible in 35 languages, and explains clearly when to access an urgent clinical opinion:

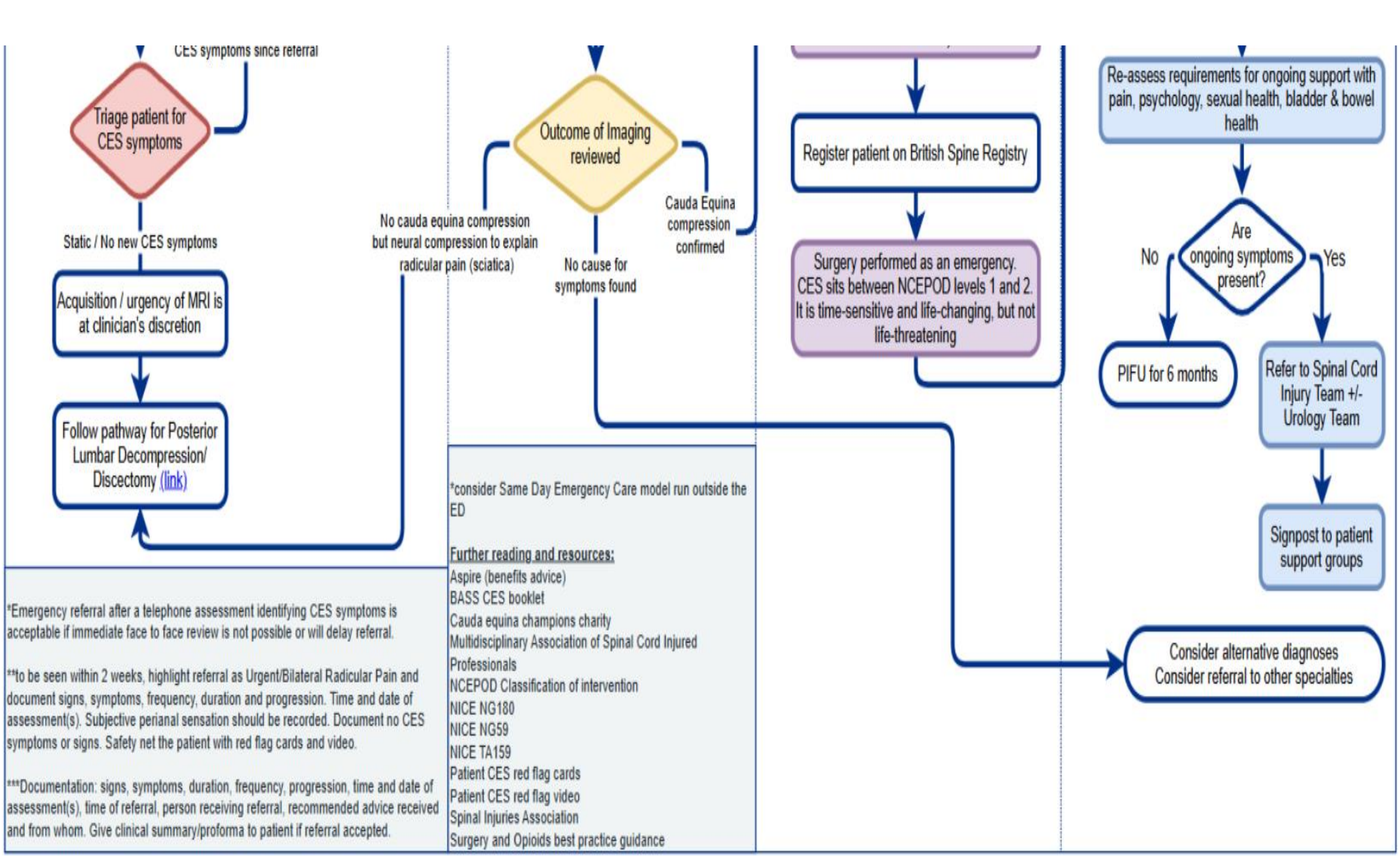


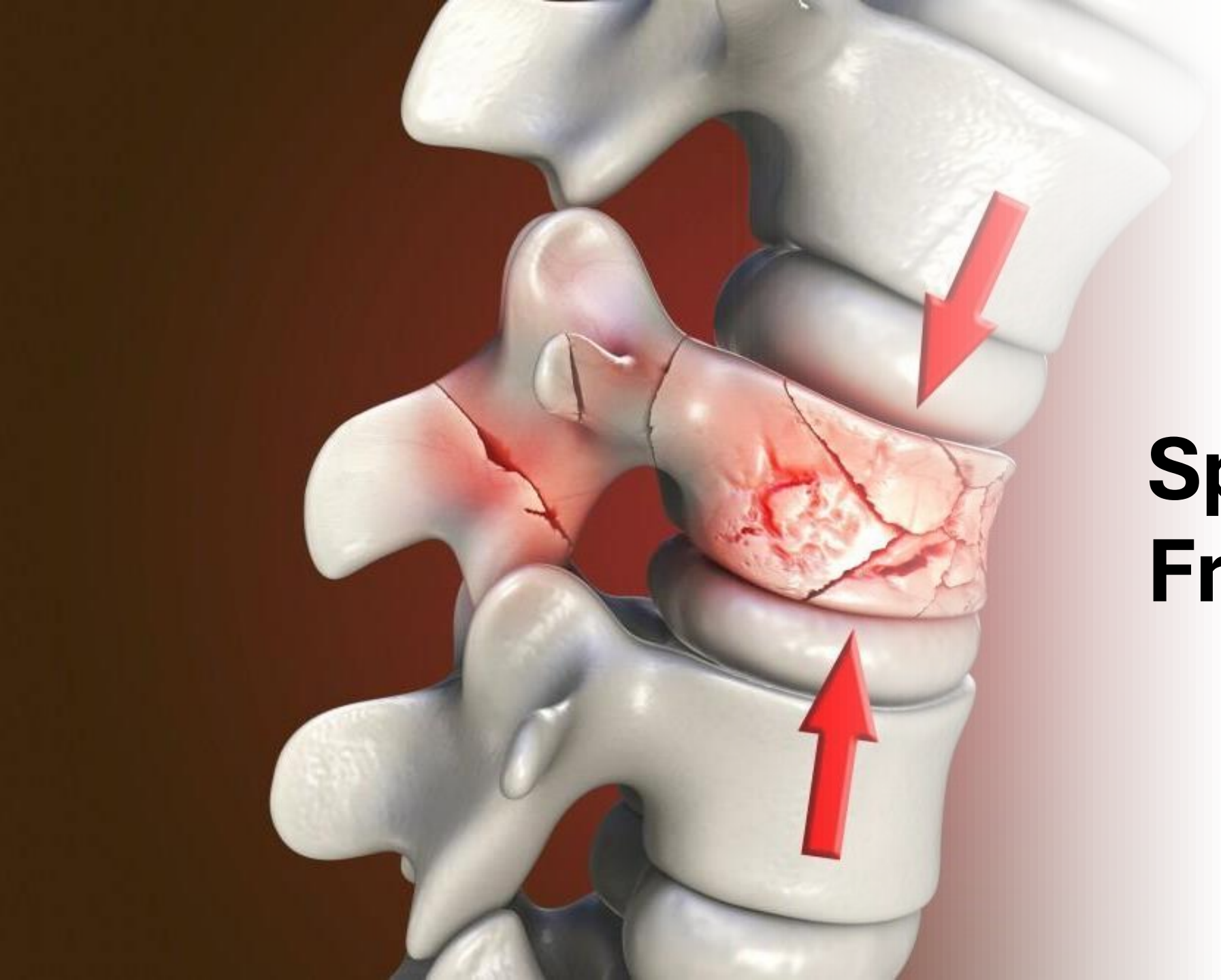
Cauda Equina Syndrome Warning Signs

- L oss of feeling/pins and needles between your inner thighs or genitals
- Numbness in or around your back passage or buttocks
- Altered feeling when using toilet paper to wipe yourself
- Increasing difficulty when you try to urinate
- Increasing difficulty when you try to stop or control your flow of urine
- Loss of sensation when you pass urine
- Leaking urine or recent need to use pads
- Not knowing when your bladder is either full or empty
- Inability to stop a bowel movement or leaking
- Loss of sensation when you pass a bowel motion
- Change in ability to achieve an erection or ejaculate
- Loss of sensation in genitals during sexual intercourse

Any combination seek help immediately


Primary & Community Care Presentation
Hospital Presentation & Diagnosis
Surgery
Post-Operative Care
Presentation in Primary & Community Care
Triage assessment of patient and symptoms*
Sudden onset bilateral radicular pain or unilateral radicular leg pain that has progressed to bilateral without CES symptoms
or
Onset of CES symptoms >14 days
Urgent referral to MSK triage service**
<2 weeks
Patient has recent onset (≤14 days) or deteriorating CES symptoms
Refer as an Emergency to nearest hospital with Emergency MRI provision***
Deteriorating / New CES symptoms since referral
Presentation in secondary care*
Triage assessment of patient and symptoms
Patient has recent (≤14 days) onset of CES symptoms
Perform bladder scan
Emergency MRI as soon as possible, and certainly within 4hrs of request
Keep patient nil by mouth
Immediate emergency referral to spinal surgical service
If transfer required, arrange blue-light Category 2 ambulance transfer
Assessment and documentation by surgical team
Consent
Catheterisation prior to surgery (if not catheterised)
TWOC post-operatively (as soon as possible) and obtain pre/post void bladder scan
Patient mobilised (usually day 1) following surgery
Information booklet provided during inpatient stay.
Surgical, medical and physiotherapy assessments made.
Splints/orthotics fitted if required.
Assess requirements for ongoing support with pain, psychology, sexual health, bladder & bowel health
6/8 week post-surgical face-to-face outpatient review


CES symptoms since referral
Triage patient for CES symptoms
Static / No new CES symptoms
Acquisition / urgency of MRI is at clinician's discretion
Follow pathway for Posterior Lumbar Decompression/ Discectomy (link)
Outcome of Imaging reviewed
No cauda equina compression but neural compression to explain radicular pain (sciatica)
Cauda Equina compression confirmed
No cause for symptoms found
Register patient on British Spine Registry
Surgery performed as an emergency. CES sits between NCEPOD levels 1 and 2. It is time-sensitive and life-changing, but not life-threatening
Re-assess requirements for ongoing support with pain, psychology, sexual health, bladder & bowel health
Are ongoing symptoms present?
No
Yes
PIFU for 6 months
Refer to Spinal Cord Injury Team +/- Urology Team
Signpost to patient support groups
Consider alternative diagnoses
Consider referral to other specialties
*Emergency referral after a telephone assessment identifying CES symptoms is acceptable if immediate face to face review is not possible or will delay referral.
**to be seen within 2 weeks, highlight referral as Urgent/Bilateral Radicular Pain and document signs, symptoms, frequency, duration and progression. Time and date of assessment(s). Subjective perianal sensation should be recorded. Document no CES symptoms or signs. Safety net the patient with red flag cards and video.
***Documentation: signs, symptoms, duration, frequency, progression, time and date of assessment(s), time of referral, person receiving referral, recommended advice received and from whom. Give clinical summary/proforma to patient if referral accepted.
*consider Same Day Emergency Care model run outside the ED
Further reading and resources:
Aspire (benefits advice)
BASS CES booklet
Cauda equina champions charity
Multidisciplinary Association of Spinal Cord Injured Professionals
NCEPOD Classification of intervention
NICE NG180
NICE NG59
NICE TA159
Patient CES red flag cards
Patient CES red flag video
Spinal Injuries Association
Surgery and Opioids best practice guidance



Spinal Fracture

2. Spinal Fracture

- Most common serious pathology to affect spine
- 0.7%-4.5% of LBP in primary care
- 6.5% of LBP in emergencies (A&E)
- Most common in thoracic (70%) then lumbar (20%) then cervical (10%) spine

Why identify?

- Some fractures stable and for conservative mx
- Others may be complex
- May identify underlying pathology eg osteoporosis

Vertebral Fractures

- There is an increasing number of younger people are affected by insufficiency spinal fractures (fractures caused by normal stresses on weakened bone),due to a range of risk factors.
- These include:
- Excessive alcohol consumption(risk increases when drinking greater than 3 units per day), vitamin D deficiency
- Long-term corticosteroid use (greater than 5 or 7.5 mg per day over a 3-monthperiod),
- Rheumatoid arthritis,
- Diabetes,
- Smoking (greater than 20 cigarettes perday61),
- Dietary restriction,
- Eating disorders,
- and absorption problems from the gut (eg, Crohn’s disease).

Clinical Picture

People with spinal fracture commonly present with:

- Sudden onset of pain,
- Mostly located in the thoracolumbar region,
- Following low-impact trauma such as a slip or trip or lifting something while in a flexed position.
- The pain varies in presentation but is often severe and mostly localized to the area of the fracture.
- Weight-bearing activities and active movements are restricted and painful,
- And the person may require strong analgesia, particularly in the early stages.

On physical examination:

- The person may have an increased prominence of the spinous process at the affected level, and increased kyphosis.
- The person may be tender to percussion at the affected level, though absence of this should not reassure the clinician that there is no fracture.

People with a suspected fracture should have an X-ray in the first instance to determine whether a fracture is present, and to grade and de-fine the nature of the fracture.

Differential Diagnosis

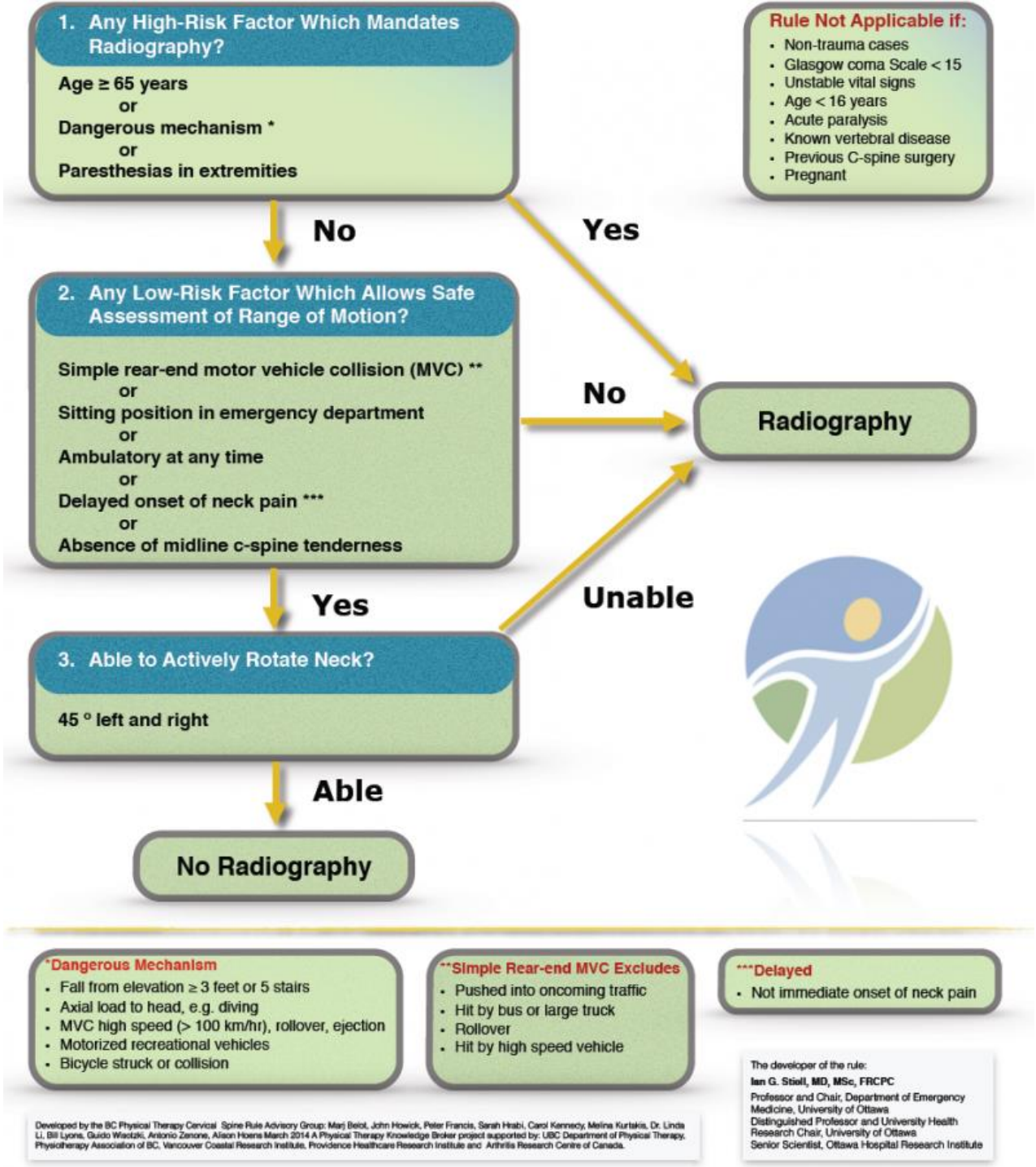
- Consider possible differential diagnoses for spinal fracture.
- metastatic spinal disease and multiple myeloma (both of which can cause healthy bone to be replaced by tumour).
- In the case of metastatic disease,60% of metastases occur in the anterior half of the vertebral body, thus potentially weakening this area and leading to a wedge fracture.
- These fractures may look very similar on X-ray.
- Fractures from myeloma may also look very similar to osteoporotic fractures on X-ray, depending on the location.
- People with myeloma may present at a slightly earlier age than those with osteoporosis and metastases, but further imaging maybe required to establish the cause of a fracture if there are no clear indications from the person’s subjective history.

cervical spine injury

Assess whether the person is at high, low or no risk for cervical spine injury using the Canadian C-spine rule as follows:

the person is at high risk if they have at least one of the following high-risk factors:

- age 65 years or older
- dangerous mechanism of injury (fall from a height of greater than 1 metre or 5 steps, axial load to the head – for example diving, high-speed motor vehicle collision, rollover motor accident, ejection from a motor vehicle, accident involving motorised recreational vehicles, bicycle collision, horse riding accidents)
- paraesthesia in the upper or lower limbs



the person is at low risk if they have at least one of the following low-risk factors:

- involved in a minor rear-end motor vehicle collision
- comfortable in a sitting position
- ambulatory at any time since the injury
- no midline cervical spine tenderness
- delayed onset of neck pain
- the person remains at low risk if they are:
- unable to actively rotate their neck 45 degrees to the left and right (the range of the neck can only be assessed safely if the person is at low risk and there are no high-risk factors).
- the person has no risk if they:
- have one of the above low-risk factors and
- are able to actively rotate their neck 45 degrees to the left and right.
- Be aware that applying the Canadian C-spine rule to children is difficult and the child's developmental stage should be taken into account.

Assessment for thoracic or lumbosacral spine injury

Assess the person with suspected thoracic or lumbosacral spine injury using these factors:

- age 65 years or older and reported pain in the thoracic or lumbosacral spine
- dangerous mechanism of injury (fall from a height of greater than 3 metres, axial load to the head or base of the spine – for example falls landing on feet or buttocks, high-speed motor vehicle collision, rollover motor accident, lap belt restraint only, ejection from a motor vehicle, accident involving motorised recreational vehicles, bicycle collision, horse riding accidents)
- pre-existing spinal pathology, or known or at risk of osteoporosis – for example steroid use
- suspected spinal fracture in another region of the spine
- abnormal neurological symptoms (paraesthesia or weakness or numbness)

on examination:

- abnormal neurological signs (motor or sensory deficit)
- new deformity or bony midline tenderness (on palpation)
- bony midline tenderness (on percussion)
- midline or spinal pain (on coughing)
- on mobilisation (sit, stand, step, assess walking): pain or abnormal neurological symptoms (stop if this occurs).

Spinal Fracture Red Flags

Hx of osteoporosis	Corticosteroid use	Previous hx of cancer	Severe trauma	Female sex
Age	Previous spinal fracture	Hx of falls	Severe new pain	Thoracic pain
Neurological symptoms	Vitamin D deficiency	Smoking > 20/day	Excessive alcohol intake	PMHx RA, Diabetes, Crohns
		Dietary insufficiency/Eating disorder		

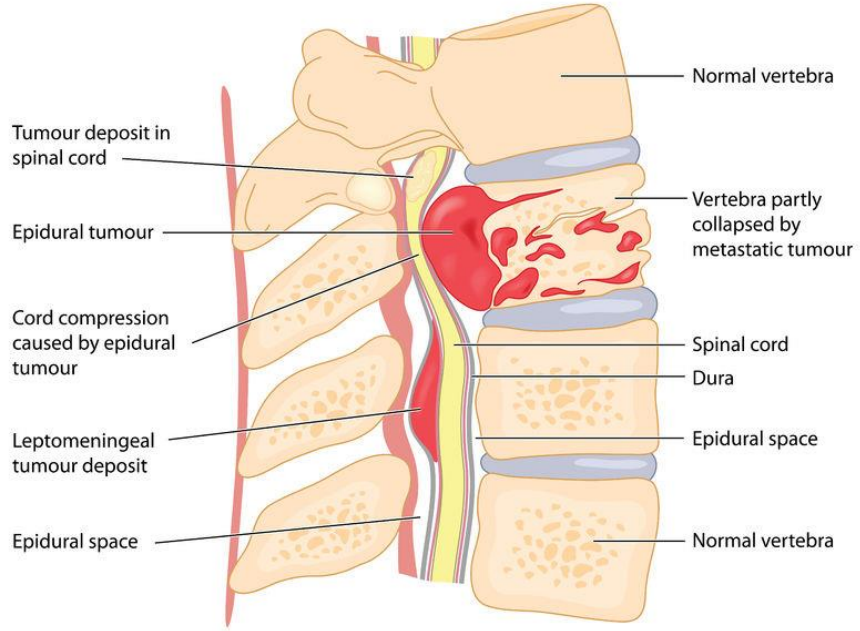
Spinal Fracture

- MRI, CT or X-Ray: NG41



Spinal Malignancy

3. Spinal Malignancy



- Metastases are cancer lesions that havespread from the primary cancer site to anew and different site in the body.
- Second most common pathology to affect spine
- 0.7% of LBP in primary care setting
- Metastases are cancer lesions that spread from primary site
- The 5 most common cancers to metastasize are breast, prostate, lung, kidney, and thyroid.
- Metastatic spinal cord compression (MSCC) – when vertebral body collapse or tumour compresses spinal cord
- MSCC first sign of mets in 25% of people without primary cancer diagnosis
- Mets most common in thoracic (70%) then lumbar (20%) then cervical (10%) spine

Clinical Picture

- The spine is one of the earliest sites affected by metastatic bone disease, especially in those cancers that have a propensity to metastasize.
- Approximately 30% of all people with one of these primary diagnoses of cancer will have their cancer metastasize, so it is important not to subject all people with a history of cancer to unnecessary and worrying investigations.
- In breast cancer, MBD can occur at any time, with 50% occurring within the first 5 years after a primary diagnosis of cancer and the other 50% developing 10 years and later.
- Other primary cancers may metastasize, but have a lower incidence.

Spinal Malignancy Red Flags

Hx of cancer	Severe, progressive, constant pain	Unremitting night pain
Systemically unwell	Night sweats	Thoracic pain
Neurological symptoms	Unexplained weight loss	Unfamiliar back pain


Spinal Infection

4. Spinal Infection

- Spinal infection is an infectious disease that affects the spinal structures, including the vertebrae, intervertebral discs,and adjacent paraspinal tissues.
- In high-income and upper middle–income countries, spinal infection has steadily increased over recent years, possibly due to an aging population and an increase in intravenous drug abuse.
- In lower middle–income and lower-income countries, spinal infection has increased due to the dual epidemic of HIV/AIDS and tuberculosis (TB).
- TB, discitis, spinal abscesses
- Uncommon – 0.01% of all non mechanical LBP in primary care
- Spinal infection represents 2% to 7% of all musculoskeletal infections.
- Discitis mostly affects lumbar (58%) then thoracic (30%) then cervical (11%) spine

Clinical Picture

- In cases of spinal infection, the time between onset and diagnosis is often prolonged.
- People can remain relatively healthy until symptoms manifest in the later stages of the disease.
- Unlike malignancy, where symptoms wax and wane, spinal infection has a more linear progression, with back pain being the most common presenting symptom, which can progress to neurological symptoms.
- If not treated in a timely manner, the condition can progress with serious complications such as paralysis or instability of the spine, and can ultimately be fatal.

Spinal Infection Red Flags

Immunosuppression	Invasive surgery	IV Drug Use	Hx of TB
Social/environmental factors	Recent pre-existing infection	Neurological symptoms	Fatigue
Fever	Night sweats	Unexplained weight loss	Non mechanical severe pain

Diagnosis

- The subjective history should consider determinants that can be divided into comorbidities, environmental factors, and social factors.
- Comorbidities that suppress a patient's immune system, such as diabetes, HIV, long-term steroid use, and smoking, put the person at risk of infection.
- Consider social and environmental factors like intravenous drug use, obesity, birth in a TB-endemic country, family history of TB, and living conditions (overcrowded living, home-lessness, imprisonment, or rural environment).
- Spinal surgery is a key risk factor for spinal infection, in particular multiple revision surgery of the lumbar spine, with an added increased risk for obese people.
- The classic triad of clinical features comprises back pain, fever, and neurological dysfunction. However, many people do not present with all 3 features.
- Only 50% of people report fever as a symptom.
- Absence of fever cannot rule out spinal infection.

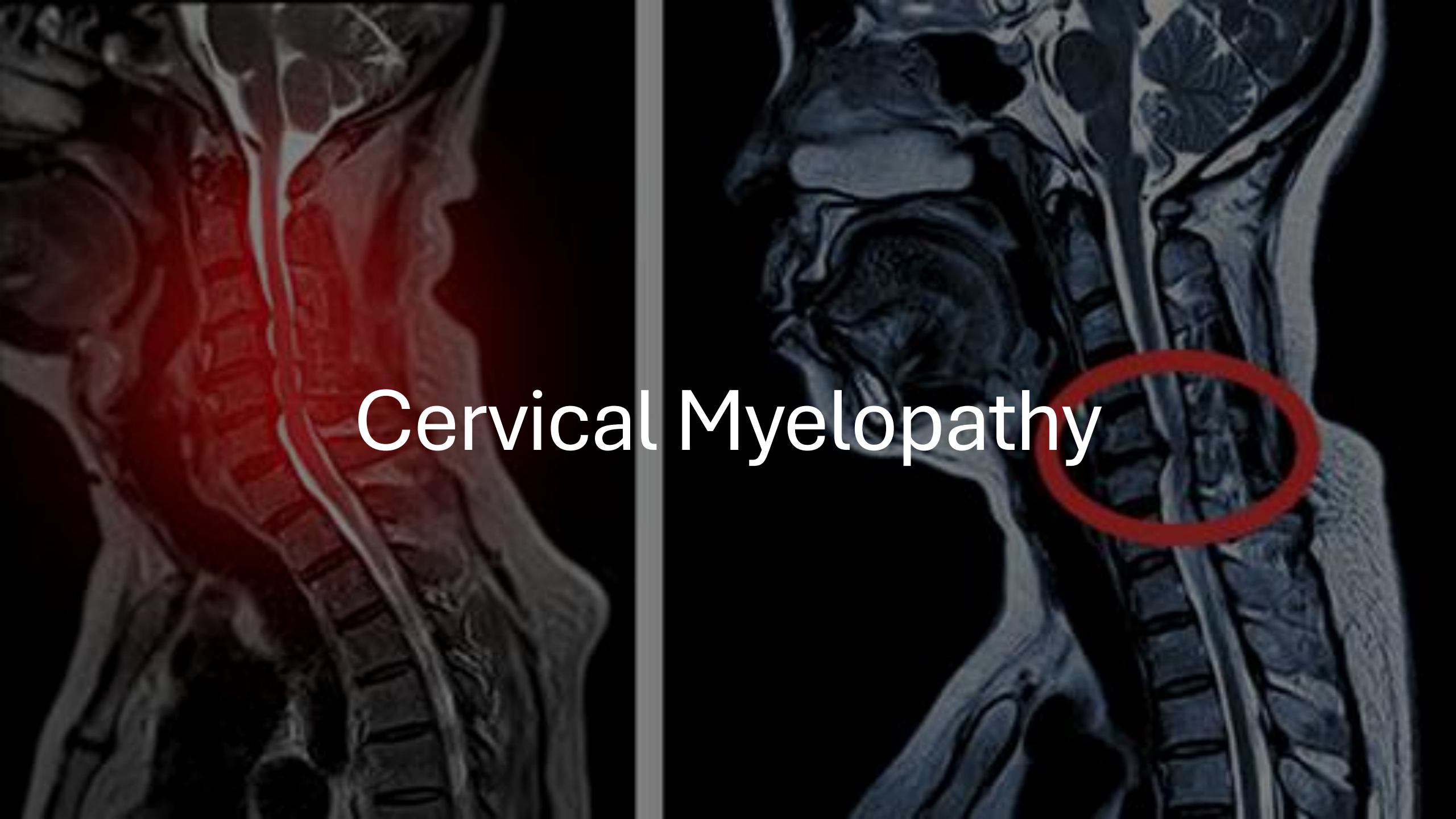
Spinal Infection

EXAMINATION

- Usual spinal ax
- May require neuro exam
- May need wider systems check eg vitals, weight check, resp ax etc.

MANAGEMENT

- MRI
- Bloods
- CXR if resp symptoms
- Urgent letter to GP for further mx


Cervical Myelopathy

6. Cervical Myelopathy

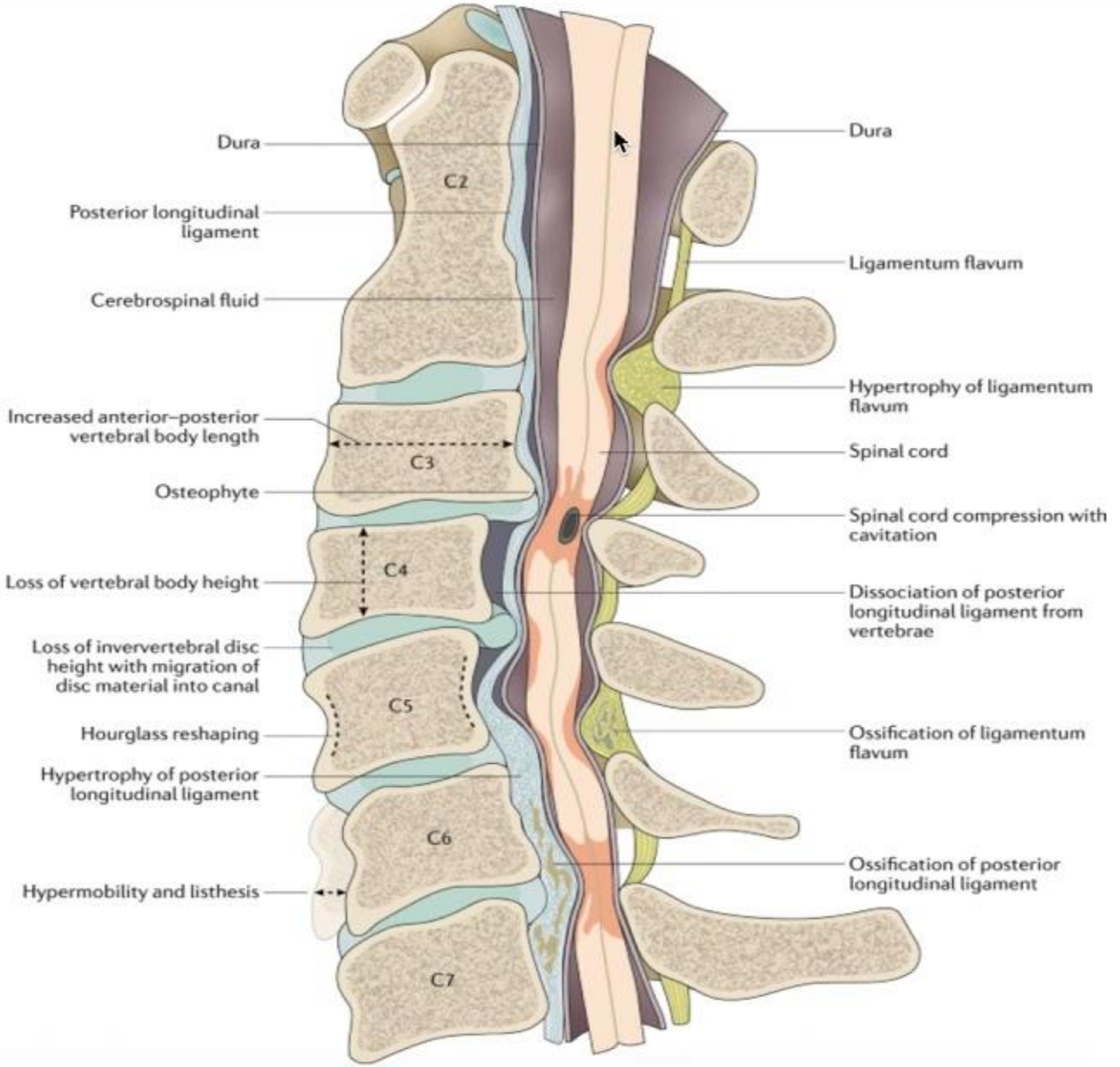
Cervical Myelopathy

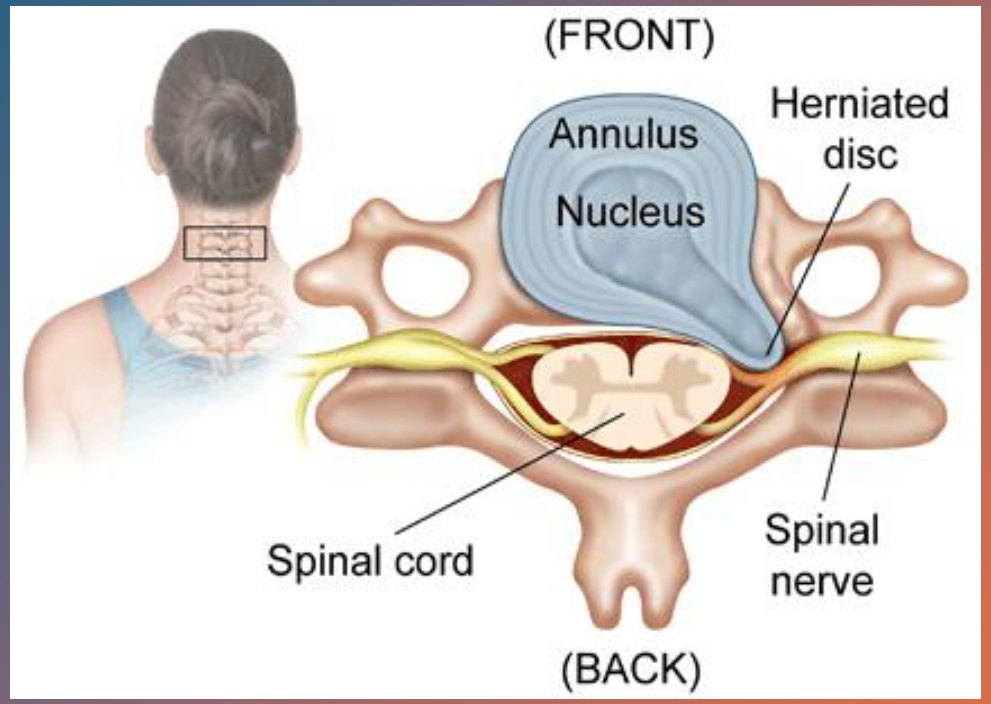
Description | impingement on the spinal cord due to a vertebral canal obstruction in the C-spine

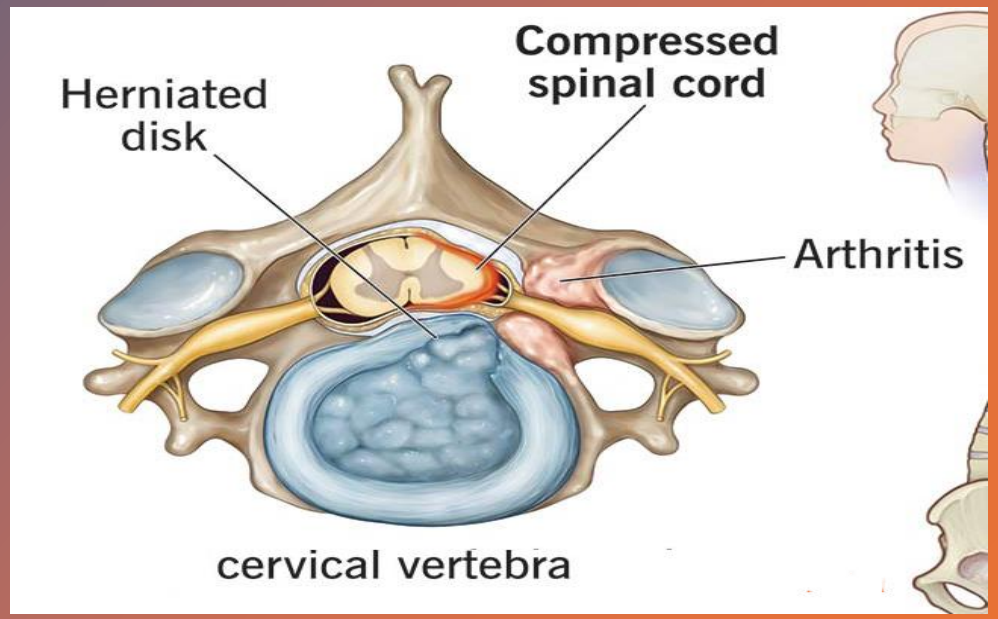
- **Risk Factors - Anything that narrows vertebral canal**
  - Cervical spondylosis
  - Age ≥ 50-60
  - Neck trauma
    - MVA
    - Sports injury
  - RA (C1-C2 interval)

S/S

- Neck and upper extremity pain
- Weakness and sensory impairments
  - UE > LE
- LMN signs at the level of lesion (weakness and hyporeflexia)
- UMN signs below the level of lesion (spasticity & hyperreflexia)
  - UMN Signs (Hoffman's, clonus, and Babinski reflexes)
- Parasthesia with weakness and wasting of the hands
- Gait disorders (wide based, ataxia)
- Bowel and bladder dysfunction
- Loss of deep touch, vibration, and joint position sense (posterior column involvement)
- Lhermitte's sign







Cervical myelopathy

- Cervical myelopathy is spinal cord dysfunction due to compression caused by narrowing of the spinal canal.
- Common causes include disc herniation, spondylosis, and congenital stenosis, often in combination.
- The compression causes upper and lower motor and sensory neurone symptoms of the arms and legs, and the onset is often insidious.

Diagnosis

- The presenting features of cervical myelopathy (cord compression) are often non-specific symptoms such as:
    - Clumsiness of hands and feet,
    - Decreased manual dexterity,
    - unsteady gait.
    - Cervical pain may be present but its absence does not exclude this diagnosis.
    - Difficulty undertaking common everyday tasks such as holding a cup or climbing the stairs.
    - These difficulties can be associated with symptoms such as numbness and tingling in the hands and feet.
    - Asking about urinary and bowel function is essential as alterations of these may be an indication of severe cord compression.

Examination

- Examination is important in cervical myelopathy and is key to suspecting the diagnosis.
- Both the upper and lower limbs should be examined as cervical myelopathy causes upper motor neurone signs, particularly in the legs (brisk reflexes and Babinski reflex).
- Weakness can be difficult to detect but clonus (more than 3 beats) is highly suggestive of cervical myelopathy.
- Romberg's test may be positive, especially in more severe cases: the patient become unbalanced when standing with arms stretched forward and eyes closed.
- Toe-heel walking is difficult in cervical myelopathy.
- Inverted Pronator Sign
- Pronator Drift







Cervical Myelopathy Red Flags

- Progressive unilateral / bilateral UL/LL pain
- Progressive unilateral / bilateral UL/LL sensory disturbance
- Age > 45
- Wasting of hand
- Hand paraesthesia
- Clumsy hands / dropping things / loss of dexterity
- Ataxic wide based gait
- Frequent falls
- CES red flags
- May include cancer red flags

Cervical Myelopathy MANAGEMENT



MRI



Onward referral

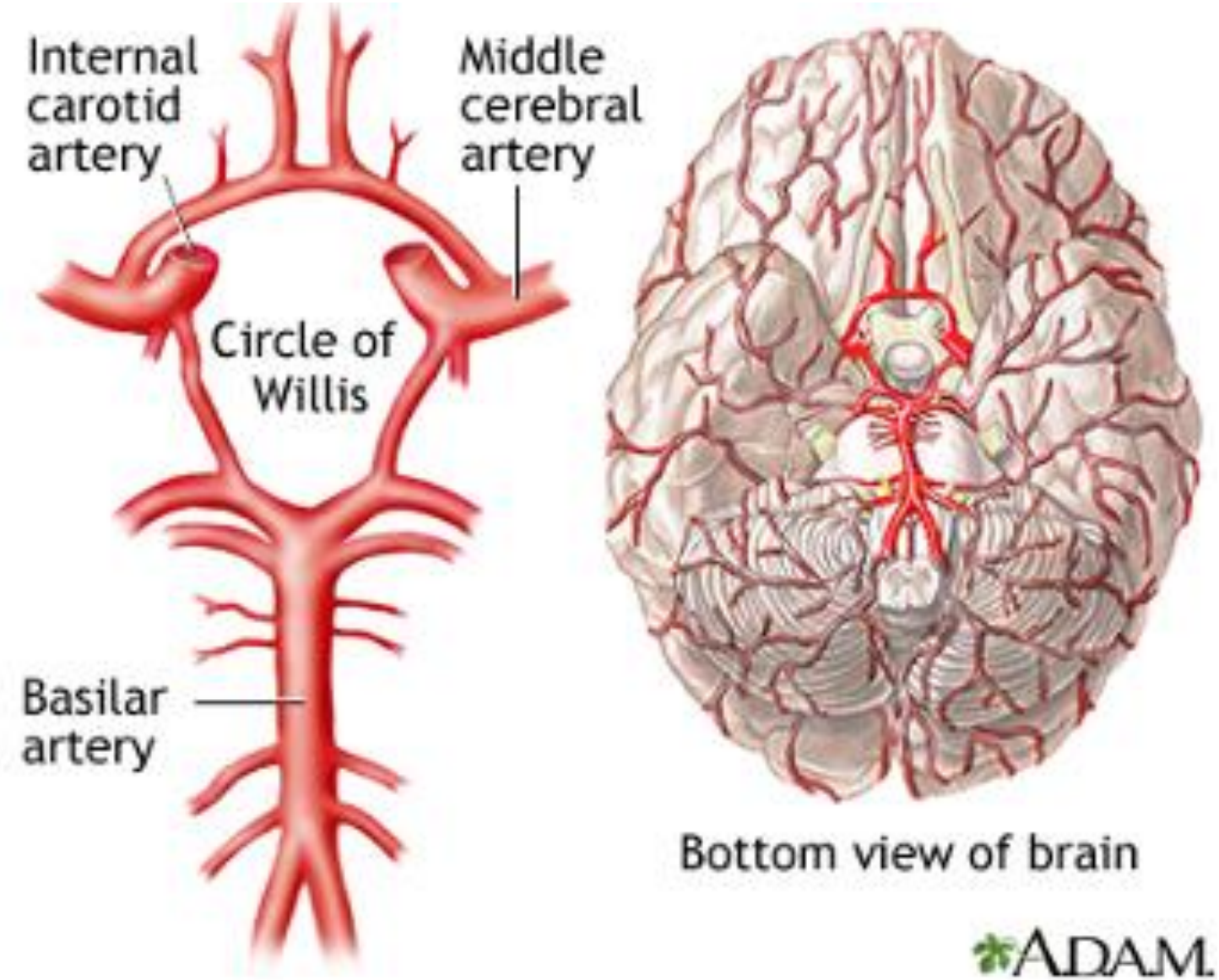


Conservative or Surgical



Urgent letter to GP for further mx

Cervical arterial dysfunction (CAD)

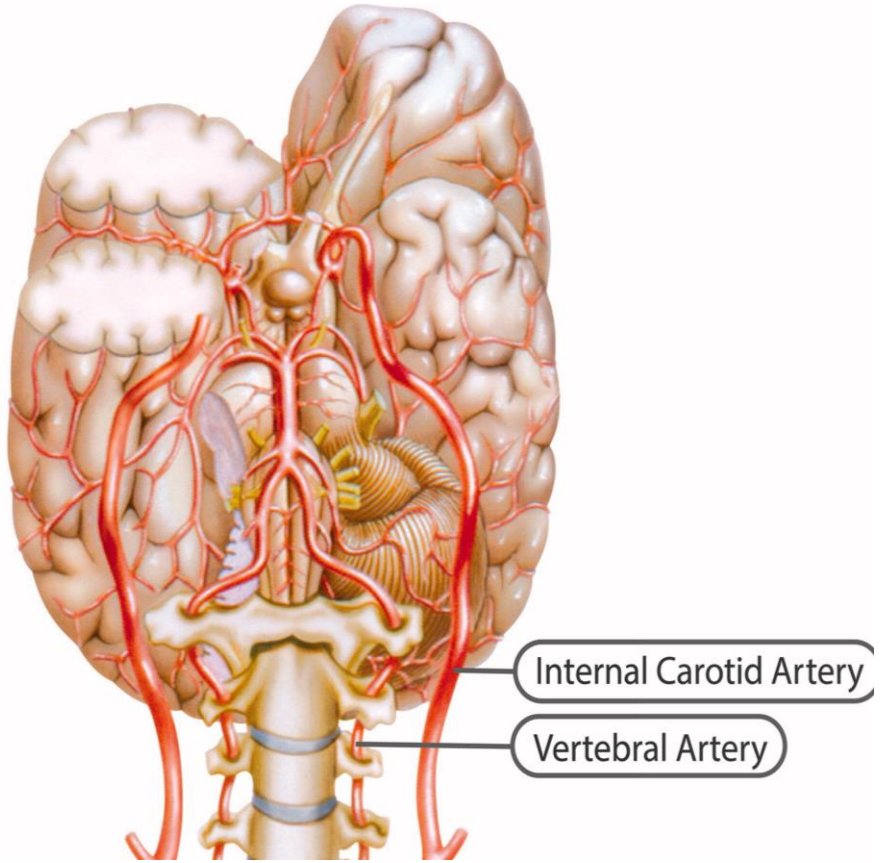


Cervical arterial dysfunction (CAD)

- CAD is an umbrella term used in manual therapy and physiotherapy to cover a range of vascular pathologies which may lead to cervicocranial ischaemia.
- CAD is inclusive of all known vascular pathologies and anatomical structures that may be compromised by movement or manual therapy.

Internal Carotid Artery Knowledge of the anterior system is important for physiotherapists because:

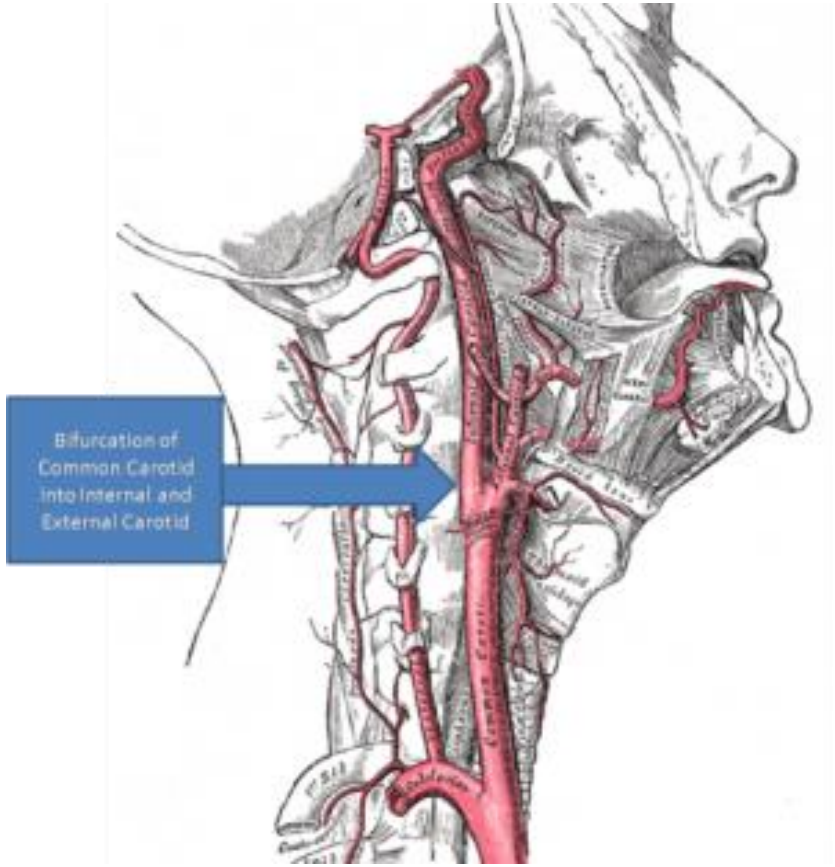
1. The internal carotid artery (ICA) provides the most significant proportion of blood to the brain.
2. Pathological changes of the ICA leading to stenosis are very common.
3. Blood flow in the ICA is known to be influenced by movement of the neck.
4. The ICA is commonly implicated in ischaemic stroke



External Carotid Artery (ECA)

The ECA is a major artery of the cranio-cervical region.

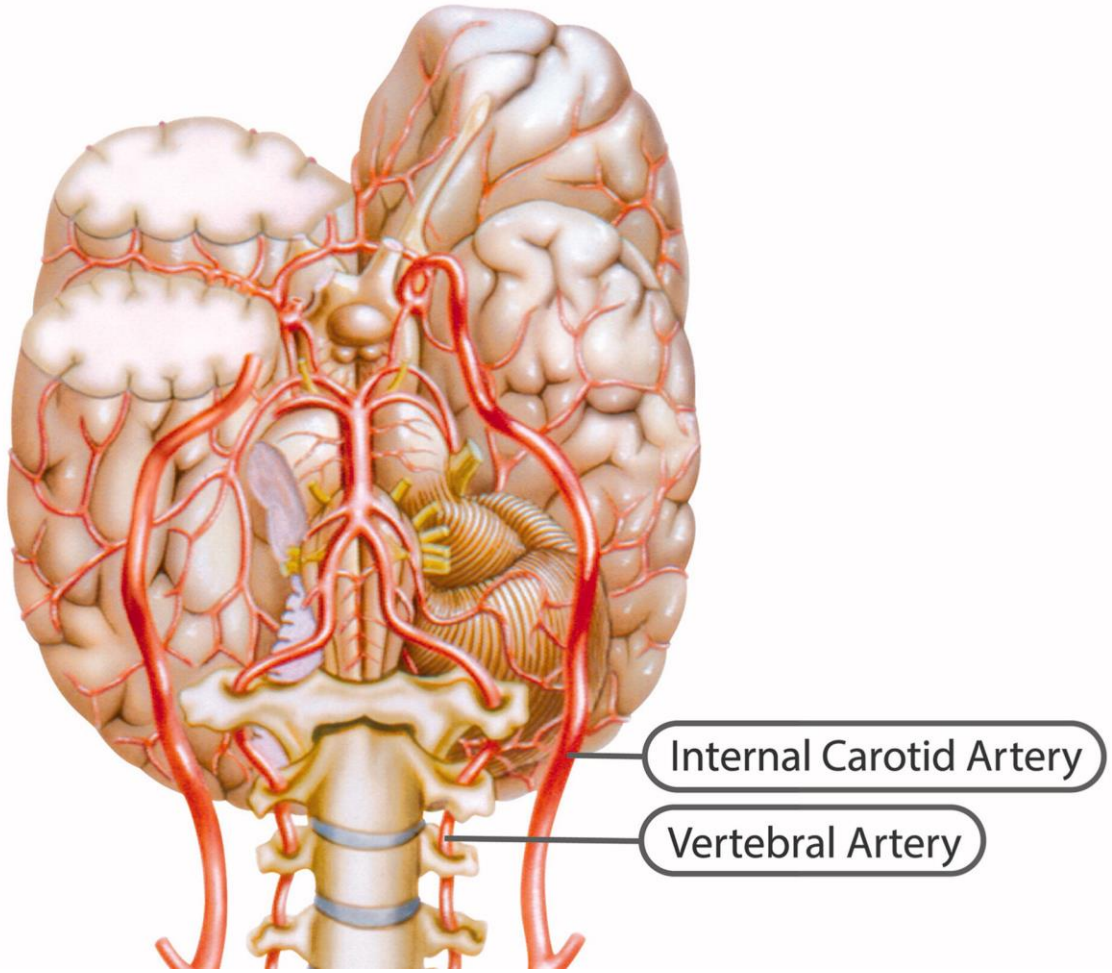
- It arises from the common carotid artery at a bifurcation point around the C3 level (upper thyroid cartilage).
- It eventually forms the occipital artery and terminates in the maxillary artery and superficial temporal artery which may be implicated in cases of headache linked to temporal arteritis.



Vascular Pathologies

- Why are we concerned?

1. Risk of affecting blood supply with manual therap
2. Vascular pathology masquerading as MSK



vertebrobasilar insufficiency (VBI)

- The VBA system supplies circulation to the hind-brain (i.e. brain-stem, medulla oblongata, pons, cerebellum, and vestibular apparatus). Entry to the skull is via the foramen magnum of the occiput.

Risk Factors of VBI

- Past history of trauma to cervical spine / cervical vessels
- History of migraine-type headache
- Hypertension
- Hypercholesterolemia / hyperlipidemia
- Cardiac disease, vascular disease, previous cerebrovascular accident or transient ischaemic attack
- Diabetes mellitus
- Blood clotting disorders / alterations in blood properties (e.g. hyperhomocysteinemia)
- Anticoagulant therapy
- Long-term use of steroids
- History of smoking
- Recent infection
- Immediately post-partum
- Trivial head or neck trauma
- Absence of a plausible mechanical explanation for the patient’s symptoms.

7. Cervical arterial dysfunction (CAD)

Structure/site	Pathology	Symptoms/Presentation
Carotid artery	Atherosclerosis Stenotic Thrombotic Aneurysmal	Carotidynia[3], neck pain, facial pain, headache, cranial nerve dysfunction, Horner's Syndrome, transient ischaemic attack (TIA), stroke
Carotid artery	Hypoplasia	Commonly silent, rare cerebral ischaemia
Carotid artery	Dissection	Neck pain, facial pain, headache, TIA, cranial nerve palsies, Horner's syndrome
Vertebral artery	Atherosclerosis	Neck pain, occipital headache, possible transient ischaemic attack (TIA), stroke
Vertebral artery	Hypoplasia	Commonly silent, rare cerebral ischaemia
Vertebral artery	Dissection	Neck pain, occipital headache, TIA, cranial nerve palsy
Temporal/ Vertebral/ Occipital/Carotid arteries	Giant cell arteritis	Temporal pain (headache), scalp tenderness, jaw and tongue claudication, visual symptoms (diplopia or vision loss – may be permanent)
Cerebral vessels	Reversible cerebral vasoconstriction syndrome (RCVS)	Severe 'thunderclap' headaches
Subarachnoid	Heamorrhage	Sudden severe headache, stiff neck, visual disturbance, photophobia, slurred speech, sickness, unilateral weakness,
Jugular vein	Thrombosis	Neck pain, headaches, fever, swelling around neck/angle of jaw
Any other cervico-cranial vessel	Vascular anomaly or malformation	Possible headache/neck pain i.e. un-ruptured carotid aneurysm

Risk

- The rate of vertebral artery (VA) dissections in gen pop 0.75-2.9 per 100,000.
- Prevalence of VA dissections associated with manual therapy approz 0.4-5 per 100,000 patients
- Relative risk of stroke following manual therapy is thought to vary between 0.14 and 6.66.
- (Nielsen et al, 2017).

Patient History

Head and/or neck pain can be a symptom of an underlying vascular pathology or dysfunction

CONSIDER

Is there presence of frank vascular pathologies of neck?
Subtle signs and symptoms of the suspected pathologies should be recognised
Is there predisposition to vascular pathologies of the neck?
Risk factors indicating the potential for neuro-vascular pathology should be recognised

DISSECTING STROKE

Recent trauma may represent an important significant risk factor for dissection

RISK FACTORS	SYMPTOMS	SIGNS
• Recent trauma • Vascular anomaly • Current or past smoker • Etc...	• Headache • Neck pain • Visual disturbance • Paresthesia (upper limb, face, lower limb) • Dizziness • Etc...	• Unsteadiness • Ptosis • Weakness (upper limb,lower limb) • Facial palsy • Speech difficulties • Swallowing difficulties • Nausea/vomiting • Dizziness • Drowsiness • Loss of consciousness • Confusion • Etc...

NON-DISSECTING STROKE

Cardiovascular risk factors are more common in older patients for artherosclerotic (disease) events

RISK FACTORS	SYMPTOMS	SIGNS
• Current or past smoker • Hypertension • High cholesterol • Etc...	• Headache • Paresthesia (upper limb, lower limb, face) • Visual disturbance • Neck pain • Dizziness • Etc...	• Weakness (upper limb,lower limb) • Speech difficulties • Ptosis • Facial palsy • Unsteadiness • Confusion • Vomiting • Swallowing difficulties • Loss of consciousness • Drowsiness • Etc...

Look out for these features:

- Older patients over 65 years
- Reports of any of the following typical symptoms of VBI

5 Ds

1. Dizziness and/or unsteadiness
2. Diplopia – double vision, visual field loss
3. Dysarthria/dysphasia – difficulty with speech or finding words
4. Dysphagia – difficulty swallowing or unexplained hoarse voice
5. Drop attacks – sudden collapse without loss of consciousness



3 Ns

1. Nystagmus (spontaneous)
2. Nausea/vomiting
3. Numbness or paraesthesia (perioral)



Physical Examination

No single test alone will provide decision-making information
Positional testing is unlikely to influence decision making

Clinical suspicion of vascular causes is supported by reasoned historical and or clinical examination findings

Are there any precautions or contraindications to physical examination?

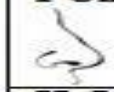
Conduct Appropriate Elements of the Physical Examination:

- *Neurological examinantion (cranial and peripheral nerves)*
- *Co-ordination and gait consideration*
- *Blood pressure*
- *Auscultation*



Refer for further vascular investigation

Risk and benefit analysis

Nerve	Classification	Major functions	Assessment
I Olfactory 	Sensory	Smell	Have patient identify a familiar scent with eyes closed (usually deferred).
II Optic 	Sensory	Vision (acuity and field of vision); pupil reactivity to light and accommodation (afferent impulse)	Have patient read from a card or newspaper, one eye at a time. Test visual fields by having patient cover one eye, focus on your nose, and identify the number of fingers you're holding up in each of four visual quadrants.
III Oculomotor 	Motor	Eyelid elevation; most EOMs; pupil size and reactivity (efferent impulse)	Check pupillary responses by shining a bright light on one pupil; both pupils should constrict. Do the same for other eye. To check accommodation, move your finger toward the patient's nose; the pupils should constrict and converge. Check EOMs by having patient look up, down, laterally, and diagonally.
IV Trochlear 	Motor	EOM (turns eye downward and laterally)	Have patient look down and in.
V Trigeminal 	Both	Chewing; facial and mouth sensation; corneal reflex (sensory)	Ask patient to hold the mouth open while you try to close it and to move the jaw laterally against your hand. With patient's eyes closed, touch her face with cotton and have her identify the area touched. In comatose patients, brush the cornea with a wisp of cotton; the patient should blink.
VI Abducens 	Motor	EOM (turns eye laterally)	Have patient move the eyes from side to side.
VII Facial 	Both	Facial expression; taste; corneal reflex (motor); eyelid and lip closure	Ask patient to smile, raise eyebrows, and keep eyes and lips closed while you try to open them. Have patient identify salt or sugar placed on the tongue (usually deferred).
VIII Acoustic/ Vestibularcochlear 	Sensory	Hearing; equilibrium	To test hearing, use tuning fork or rub your fingers, place a ticking watch, or whisper near each ear. Equilibrium testing is usually deferred.
IX Glossopharyngeal 	Both	Gagging and swallowing (sensory); taste	Touch back of throat with sterile tongue depressor or cotton-tipped applicator. Have patient swallow.
X Vagus 	Both	Gagging and swallowing (motor); speech (phonation)	Assess gag and swallowing with CN IX. Assess vocal quality.
XI Spinal accessory 	Motor	Shoulder movement; head rotation	Have patient shrug shoulders and turn head from side to side (not routinely tested).
XII Hypoglossal 	Motor	Tongue movement; speech (articulation)	Have patient stick out tongue and move it internally from cheek to cheek. Assess articulation.

Risks & Benefits of Orthopaedic Manipulative Therapy

Person with neck and/or head pain

Vasculogenic hypothesis excluded

Person-centred shared decision-making

Risk

- Very Low absolute risk 0.006%
- Less Risk compared to alternative treatments from NSAIDS and analgesics 6%

Management
Risk:Benefit analysis

Reasoning empowers patient and therapist to establish risk factors for vascular pathologies (trauma, migraine, cardiovascular profile) and avoid known provocative positions (e.g. rotation and extension)

Consider Suitability of patient for OMT

Consider Alternative (non-OMT) treatment options

Benefit

- Moderate to Large effect sizes for neck pain, disability, funtional outcomes
- Moderate to large effect size for cervical radiculopathy
- Favourable outcomes for tension-type headache
- Adding OMT to exercise enhances effect

Monitor responses and progress in-line with clinical reasoning process

8. Inflammatory Pathology

- Can affect the spine [Axial Spondyloarthropathy (AxSpA)] or peripheral joints [Peripheral SpA]
- AxSpA can be nonradiographic
- Over time nonradiographic AxSpA can progress to radiographic, also known as ankylosing spondylitis
- Inflammatory back pain can also occur in psoriatic arthritis, enteropathic arthropathy, juvenile idiopathic arthritis, reactive arthritis……

INFLAMMATORY BACK PAIN

- M> F early onset <45 years of age)
- Lasts longer than three months
- Is worse after rest (eg, in the morning)
- Improves with activity
- Wakes you up in the second half of the night
- Can cause morning stiffness lasting more than 30 minutes
- Can be associated with buttock pain that alternates between the left and right sides

EXTRA ARTICULAR SIGNS

- **S**kin – Psoriasis
- **C**rohn's or Colitis/Inflammatory Bowel Disease
- **R**elatives - strong family hx HLAB27 antigen
- **E**yes – anterior uveitis
- **E**arly morning stiffness > 30 mins
- **N**ail lesions like thickening, depressions
- **D**actylitis – sausage like swelling of digits
- **E**nthesitis – Achilles, PF, patella tendon
- **M**ovement + Medication effect - NSAIDs

9. Visceral Pathology

